IN THE UNITED STATES DISTRICT COURT
FOR THE DISTRICT OF COLORADO

Civil Action No. 09-cv-00019-MSK-MEH

UNITED STATES OF AMERICA

Plaintiff,

v.

CEMEX, INC.,

Defendant.

UNITED STATES' MOTION FOR PARTIAL SUMMARY JUDGMENT

Pursuant to Federal Rule of Procedure 56(a), Plaintiff United States of America hereby moves for partial summary judgment on parts of the First and Second Claims for Relief in its First Amended Complaint (Doc. # 16), seeking a determination that the appropriate test for determining whether a "significant net emissions increase" has occurred at the CEMEX, Inc. ("CEMEX") Lyons, Colorado facility (the "Lyons Facility" or "Facility") is the "actual to potential-to-emit" test. This threshold question rests on the issue of whether CEMEX had "begun normal operations" within the meaning of the Colorado State Implementation Plan ("SIP") and federal regulations implementing the Prevention of Significant Deterioration ("PSD") and Non-attainment New Source Review ("NNSR") provisions of the Clean Air Act, 42 U.S.C. §§ 7470 - 7515, prior to completing the modification at the Lyons Facility that is at issue in this case. There is no genuine dispute of material fact that CEMEX had not "begun normal operations" within the meaning of the SIP and federal regulations. As discussed below, because the Lyons Facility had not "begun normal operations" prior to completing the modification, the

PSD and NNSR[1] requirements would be triggered based on a comparison of the Facility's "actual emissions" before the modification and its "potential emissions" after the modification. Conversely, a facility that had "begun normal operations" at the time of the modification would apply a different test, requiring a different analysis. Thus, a determination that the Lyons Facility had not "begun normal operations" prior to completion of the modification at issue will greatly simplify the scope of the evidence presented at trial, as each party will only need to present evidence on one comparison – pre-modification actual emissions to post-modification potential emissions. The Lyons Facility's post-modification actual emissions will be irrelevant, as will any actual emissions to future actual emissions comparison. It will also render irrelevant expert testimony related to post-modification actual emissions, and thereby reduce the amount and scope of expert testimony subject to Fed. R. Evid. 702 challenge.

Pursuant to D.C.COLO.LCivR 7.1 A, counsel for the United States discussed the grounds for this motion and the relief requested with counsel for the Defendant on November 10, 2011. Defendant opposes the relief requested.

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[1] Colorado air quality regulations provide that "[i]n the Denver Metropolitan PM-10 nonattainment area, any net emission increase that is significant for sulfur dioxide or nitrogen oxides shall be considered significant for PM-10." Ex. 2, Excerpt of COLO. CODE REGS. § 1001-5, Colorado Air Quality Control Commission ("CAQCC") Reg. 3, Part A, Section I.B.35.B.a. at 188 (1998). Thus, to the extent the Lyons Facility triggers PSD requirements for NOx, it will also trigger NNSR for PM-10. As the Court has previously noted, the distinction between the PSD and NNSR permitting programs does not affect the present analysis. *See* Opinion and Order Denying Motion to Dismiss (Doc. # 89) at 4 n.2. The United States therefore adopts the Court's convention of using "PSD permits" or the "PSD permitting process" to reference simultaneously both the PSD and NNSR regimes, unless otherwise specified.

BACKGROUND

As a result of concerns that existing law did not protect air quality in areas that met air quality standards, Congress amended the Clean Air Act in 1977 to establish a statutory PSD program "aimed at giving added protection to air quality," *Envtl. Def. v. Duke Energy Corp.*, 549 U.S. 561, 567 (2007), while fostering economic growth in a manner consistent with preservation of existing clean air resources. 42 U.S.C. § 7470(3) (1977). The PSD program directly addresses the impact on ambient air quality resulting from new construction of, and modifications to, large pollutant-emitting facilities in areas of the country that are in compliance with ambient air quality standards. *Envtl. Def.*, 549 U.S. at 567-68; 42 U.S.C. §§ 7470, 7475(a)(3) (1977). Under the PSD program, "[n]o major emitting facility . . . may be constructed[2] . . . unless" various requirements are met, including obtaining a preconstruction permit setting forth emission limitations and applying the "best available control technology." 42 U.S.C. § 7475(a) (1977), § 7479(3) (1990). The term "major emitting facility" includes Portland cement plants with the potential to emit 100 tons per year or more of any air pollutant. 42 U.S.C. § 7479(1) (1990).

The PSD program can be implemented by EPA or by a state where it is part of an authorized SIP. The State of Colorado has promulgated a SIP that implements the PSD program, among other Clean Air Act requirements. EPA approved Colorado's PSD program as part of the

[2] Modifications are within the ambit of the PSD program because the Clean Air Act defines "construction" to include "modification." 42 U.S.C. § 7479(2)(C) (1990). "Modification" is defined as "any physical change in, or change in the method of operation of, a stationary source which increases the amount of any air pollutant emitted by such source or which results in the emission of any air pollutant not previously emitted." 42 U.S.C. § 7411(a)(4) (1990).

Colorado SIP on September 2, 1986. 51 Fed. Reg. 31,125. EPA approved a re-codified and amended version of the program effective February 20, 1997. 62 Fed. Reg. 2910.[3]

Like the EPA PSD regulations on which it is based, the Colorado SIP[4] requires that "any new major stationary source or major modification proposing to construct" in Colorado must obtain a permit that meets certain requirements including a control technology review (requiring the installation of "best available control technology"), a source impact analysis, and preconstruction monitoring and analysis. *See* Ex. 1, Excerpt of COLO. CODE REGS. § 1001-5, CAQCC Reg. 3, Section IV.D.3.a.(i) - (iii) and (vi) at 114-118 (1995). *See also* 40 C.F.R. § 52.21(j), (k), (m), and (o) (1997). A "major modification" is defined as any physical change in or change in the method of operation of a major stationary source that would result in a

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[3] Colorado's PSD/NNSR program for the relevant time period can be found at 5 COLO. CODE REGS. § 1001-2 (1995), § 1001-5, Reg. 3 (1995), and § 1001-5, Reg. 3 (1998). Every three years, EPA compiles the approved SIP provisions. Relevant excerpts from the November 1995 compilation are attached as Exhibit 1, and excerpts from the November 1998 compilation are attached as Exhibit 2. Exhibit 1 contains the relevant excerpts of the SIP provisions in effect prior to February 20, 1997, and Exhibit 2 contains the relevant excerpts of the SIP provisions effective after that date. Plaintiff previously submitted to the Court the complete 1995 and 1998 SIP compilations as Exhibit 1, Parts 1 through 5 (Docs. # 38-2 to 38-6) to United States' Memorandum in Opposition to CEMEX, Inc.'s Motion to Dismiss Pursuant to Fed. R. Civ. P. 12(b)(6) (Doc. # 38) and other excerpts from the 1995 and 1998 SIP compilations as Exhibits 1 and 2 of Plaintiff United States of America's Response in Opposition to Defendant CEMEX, Inc.'s Motion for Summary Judgment (Doc. # 158) ("Response"). Defendant relies upon the same regulatory provisions. *See* Defendant's Reply in Support of CEMEX, Inc.'s Motion for Summary Judgment (Doc. # 165) (relying on the 1995 and 1998 Colorado SIP compilations attached as Exhibit 1 and 2 of Plaintiff's Response (Doc. # 158). Thus, there is no dispute over the applicable SIP.

[4] This brief cites to the regulations contained in both the 1995 and 1998 SIP compilations, as the Lyons Facility modifications at issue spanned the two dates. While they are codified differently, the underlying provisions are substantially the same. This brief also cites to EPA's PSD regulations upon which the Colorado SIP is based.

significant net emissions increase of any pollutant subject to regulation under the Clean Air Act. Ex. 1, CAQCC Reg. 3, Section I.B.2. at 96; Ex. 2, Excerpt of COLO. CODE REGS. § 1001-5, CAQCC Reg. 3, Part A, Section I.B.35.B. at 188 (1998). *See also* 40 C.F.R. § 52.21(b)(2)(i) (1997). A "net emissions increase" is defined as "[a]ny increase in actual emissions from a particular physical change or change in method of operation" and any other emissions increase or decrease at the source that is contemporaneous and creditable. Ex. 1, Excerpt of COLO. CODE REGS. § 1001-2, CAQCC Common Provision Regulation, Section I.G., "net emissions increase," Parts a.(i)-(ii) at 72 (1995); Ex. 2, CAQCC Reg. 3, Part A, Section I.B.36.a.(i)-(ii) at 192-93. *See also* 40 C.F.R. § 52.21(b)(3)(i) (1997). An emissions increase is "significant" if the net increase or potential to emit is equal to or greater than 40 tons per year ("tpy") of NOx, among other pollutants. Ex. 1, CAQCC Common Provision Regulation, Section I.G., "significant," Part a. at 81; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.57.a. at 200. *See also* 40 C.F.R. § 52.21(b)(23)(i) (1997).

The Colorado SIP defines "actual emissions" as follows: "Actual emissions as of a particular date shall equal the average rate, in tons per year, at which the unit emitted the pollutant during a two-year period which precedes the particular date and is representative of normal unit operation." Ex. 1, CAQCC Common Provision Regulation, Section I.G., "actual emissions," Part a. at 57; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.1.a. at 177. *See also* 40 C.F.R. § 52.21(b)(21)(i)-(ii) (1997). In addition, "[f]or any emissions unit *which has not begun normal operations* on the particular date, *actual emissions shall equal the potential to emit of the unit on that date*." Ex. 1, CAQCC Common Provision Regulation, Section I.G., "actual emissions," Part c. at 57; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.1.c. at 178 (emphasis added). *See also* 40 C.F.R. § 52.21(b)(21)(iv) (1997). "Potential to emit" is "[t]he maximum capacity of a stationary source to emit a pollutant under its physical and operational design." Ex. 1, CAQCC Common Provision Regulation, Section I.G., "potential to emit," at 77; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.44 at 196. *See also* 40 C.F.R. § 52.21(b)(4) (1997).

<u>PORTION OF CLAIM UPON WHICH JUDGMENT IS SOUGHT</u>

A. The United States Is Entitled to Partial Summary Judgment that CEMEX Had Not "Begun Normal Operations" at the Time of the 1997 Era Modifications.

1. <u>Burden of Proof and Elements.</u>

To prevail on its Clean Air Act PSD claim, the United States must establish, by a preponderance of the evidence, a *prima facie* case that the Lyons Facility: (i) is a major source that (ii) undertook a physical change in or change in the method of operation (iii) that would result (iv) in a net increase in actual emissions of NOx (v) of 40 tons per year or more (vi) without obtaining a PSD permit. *See* CAQCC Reg. 3, Section IV.D.3.a (1995); CAQCC Reg. 3, Part B, Section IV.D.3.a (1998). Element (iv) can be stated with more precision for a source that has not begun normal operations by incorporating the definitions of "actual emissions" and "potential emissions" as follows: (iv) net increase in NOx comparing the average rate, in tons per year, that the Facility emitted during the two years prior to the modification (assuming the prior two years represented normal source operations) to the maximum capacity of the Facility to emit the pollutant after the modification under its physical and operational design. This element is often identified as "a net increase comparing actual emissions to potential emissions" or simply the "actual-to-potential to emit" test.

The United States seeks partial summary judgment that CEMEX at the time of the modification had not "begun normal operations" at the Lyons Facility as the phrase is used in the Colorado SIP and EPA regulations. As such, the appropriate comparison for determining whether the modification resulted in a net significant increase in NOx is the pre-modification actual emissions versus the post-modification potential emissions.

2. Element 1: The Lyons Facility Is a Major Source.

It is undisputed that the Lyons Facility is a major source of NOx emissions. *See* Second Amended Scheduling Order (Doc. # 101), June 28, 2010, p. 6 (Undisputed Facts). As such, it is an undisputed fact that Defendant is subject to the PSD regulations in the Colorado SIP. *See also* Ex. 3, Excerpt of Defendant CEMEX Inc.'s Second Amended Responses to United States' First Set of Requests for Admission, Response Nos. 18, 19, and 20, pp. 9-10 (admitting that Lyons facility was a "major stationary source" within the meaning of the PSD and NNSR Programs of the State of Colorado).

3. Element 2: In 1997-2000, CEMEX Modified the Lyons Facility.

A "major modification" is any physical change in or change in the method of operation of an existing air pollution unit. *See* Ex. 1, CAQCC Reg. 3, Section I.B.2. at 96; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.35.B. at 188. It is undisputed that CEMEX made physical and operational changes to the Lyons Facility over a three-year period beginning in 1997. In response to an interrogatory, CEMEX identified numerous modifications made to the Facility's Raw Mill System, Kiln System, and Finish Mill System during this time period. *See* Ex. 4, Excerpt of Defendant CEMEX Inc.'s Responses to United States' First Set of Interrogatories, Response to Interrogatory No. 5, pp. 12-17. For the Raw Mill, CEMEX admits that it added a third air compressor for material conveying (circa 1997) and increased the material transport line from 8 inches to 10 inches (circa 1998). *Id*. For the Kiln System, CEMEX admits that it increased the refractory thickness in certain zones of the preheater tower combustion chamber (no date provided[5]), injected purchased oxygen (circa 1997) and liquid oxygen from the VSA plant into the calciner (circa 1999),[6] and modified the coal pipe booster fan and corresponding burner pipe (circa 2000). *Id*. For the Finish Mill System, CEMEX installed a high efficiency separator (circa 1998). The foregoing changes constitute "modifications" as defined in 42 U.S.C. § 7411(a)(4) and the Colorado SIP. *Id*.

4. Elements 3-5: The Modifications Would Result in a Net Increase in Actual Emissions of NOx of 40 TPY or More.

The third, fourth and fifth elements (whether the modifications (iii) would result (iv) in a net increase in actual emissions of NOx (v) of 40 tpy or more) are difficult to assess independently. The United States recognizes that the impact of the modifications on emissions is presently disputed and thus not appropriate for summary adjudication. However, the question of which emission test applies to determine whether a "net emissions increase" occurred as a result of the modification is a legal one[7] and thus can be resolved on partial summary judgment.

[5] Deposition testimony indicates that the combustion chamber modifications took place in 1997. Ex. 5, Excerpt of Deposition of John Lohr, October 11, 2010, TR at 28:4-7 (discussing Ex. 6, Lyons Clinker Production History, May 17, 2000, CMXLYO00045592).

[6] *See* Ex. 6, Lyons Clinker Production History, May 17, 2000, CMXLYO00045592; Ex. 5, Excerpt of Deposition of John Lohr, October 11, 2010, TR at 32:10-33:5.

[7] CEMEX has recognized the legal nature of this inquiry. When asked to admit that various aspects of the modification contributed to an increase in the Lyons Facility's potential to emit, CEMEX objected, stating that the request "calls for a purely legal conclusion." *See, e.g.*, Ex. 3,

Moreover, a ruling on the appropriate test for determining whether there would be an increase in emissions at the time the modifications were contemplated[8] will simplify the evidence that must be presented at trial by eliminating the need to present other formulations of actual emissions comparisons.

The assessment of whether there is a "net increase of actual emissions" within the meaning of the Colorado SIP depends in part on the meaning of the phrase "actual emissions." The Colorado SIP states: "Actual emissions as of a particular date shall equal the average rate, in tons per year, at which the unit emitted the pollutant during a two-year period which precedes the particular date and is representative of normal unit operation." Ex. 1, CAQCC Common Provision Regulation, Section I.G., "actual emissions," Part a. at 57; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.1.a. at 177. However, the SIP recognizes a fundamental difficulty in predetermining the actual emissions of a source that has not yet been modified, as the actual emissions will occur after the modification is completed. The Colorado SIP thus provides – as do the federal regulations – that "[f]or any emissions unit which has not begun normal operations

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CEMEX Inc.'s Second Amended Responses to United States' First Set of Requests for Admissions, Response Nos. 134 and 136, pp. 62-63.

[8] The PSD regulations require a major source of air pollutants to obtain a PSD permit *before* the major source commences construction of a "modification" that would result in a significant net emissions increase of a pollutant. *See, e.g.*, Ex. 1, CAQCC Reg. 3, Section III.A.1. at 103 ("[N]o person shall commence construction of any stationary source or modification of a stationary source without first obtaining or having a valid permit from the Division."); Ex. 2, CAQCC Reg. 3, Part B, Section III.A.1 at 250 (same); *see also United States v. Cinergy Corp.*, 384 F. Supp. 2d 1272, 1276 (S.D. Ind. 2005) (citing *United States v. S. Ind. Gas & Elec. Co.*, 245 F. Supp. 2d 994, 998 (S.D. Ind., 2002) ("*SIGECO*")), *aff'd*, 458 F.3d 705 (7th Cir. 2006), *cert. denied*, 549 U.S. 1338 (2007); Defendant CEMEX, Inc.'s Motion for Summary Judgment ("CEMEX Motion for Summary Judgment") (Doc. # 147), pp. 5-6.

on the particular date, actual emissions shall equal the potential to emit[9] of the unit on that date." Ex. 1, CAQCC Common Provision Regulation, Section I.G., "actual emissions," Part c. at 57; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.1.c. at 178. *See also* 40 C.F.R. § 52.21(b)(21)(iv) (1997). The meaning of "begin normal operations" is therefore key. A facility that has not begun normal operations due to the modifications must calculate its post-modification actual emissions by determining its potential to emit.

Here, considering the plain language of the Colorado SIP, EPA's interpretation of identical language in the federal PSD regulations, and case law assessing what it means to have begun normal operations, it is clear that CEMEX had not begun normal operations prior to modification of its Facility.

(a) The Plain Language of the Colorado SIP Requires that Post-Change Emissions from the Lyons Facility Be Measured Using its "Potential to Emit."

The Colorado SIP provides that "[f]or any emissions unit which has not begun normal operations on the particular date, actual emissions shall equal the potential to emit of the unit on that date." Ex. 1, CAQCC Common Provision Regulation, Section I.G., "actual emissions," Part c. at 57; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.1.c. at 178. The reason for this is simple. PSD is a preconstruction program. Major sources must make determinations about applicability before the modification is undertaken. *See United States v. Ohio Edison*, 276 F. Supp. 2d 829, 865 (S.D. Ohio 2003). As EPA noted when promulgating the PSD regulations (upon which the

[9] "Potential to emit" is "[t]he maximum capacity of a stationary source to emit a pollutant under its physical and operational design." Ex. 1, CAQCC Common Provision Regulation, Section I.G., "potential to emit," at 77; Ex. 2, CAQCC Reg. 3, Part A, Section I.B.44. at 196. *See also* 40 C.F.R. § 52.21(b)(4) (1997).

Colorado SIP is based), "[w]here the emissions unit has not 'begun normal operations,' EPA regulations recognize that future actual emissions are difficult to predict and employ future 'potential' emissions as a proxy." 57 Fed. Reg. 32,314, 32,317 (col. 1) (July 21, 1992).

The modification Defendant made to the Lyons Facility involved changes to the Raw Mill, the Kiln System, and the Finish Mill. *See*, *supra*, discussion at pp. 6-7. CEMEX cannot dispute that the changes altered the underlying design and functionality of the Facility. For example, CEMEX sought a patent for the new oxygen plant, stating in the application that the "invention and its embodiments includes an apparatus and a method for improving combustion in a cement kiln system . . . [and] includes introducing oxygen into the precalciner of a cement kiln system. *See* Ex. 7, United States Patent, No. 6,488,765 B1 (Dec. 3, 2002). The modified facility did not begin "normal operations" until after the modification was completed – and after the Facility should have assessed the net change in emissions that would result from the modification. Therefore, pursuant to the plain language of the Colorado SIP, post-change emissions from the modified Lyons Facility must be measured using the Facility's "potential to emit."

(b) This Interpretation of the Colorado SIP Is Consistent with EPA's Interpretation of the Identical Language in its PSD Regulations.

EPA interprets the "potential to emit" provision within the definition of "actual emissions" in the PSD regulations – which is identical to the Colorado SIP definition – as applying to new and modified units:

> [T]he PSD regulations provide that the increase in emissions is determined by subtracting the affected units' pre-change "actual emissions" (referred to above as the "baseline") from their post-change "actual emissions." For units that have not "begun normal operations," the regulations generally provide that actual

> emissions are equal to the units' "potential to emit." *EPA interprets this provision to mean that units which have undertaken a non-routine physical or operational change have not "begun normal operations" within the meaning of the PSD regulations*, since pre-change emissions may not be indicative of how the units will be operated following the non-routine change.

Ex. 8, *In re Monroe Electric Generating Plant*, p. 15, n15 (emphasis added) (citations omitted). *See also* 45 Fed. Reg. 52,675, 52,677 (col. 2) (Aug. 7, 1980) (explaining that post-change emissions "will generally be the potential to emit of the new or modified unit"); Ex. 9, Memorandum from John Calcagni to William B. Hathaway, September 18, 1989, p. 3 (noting that where a change will affect normal operations of an existing unit "(as in the case of a change which could result in increased use of the unit), 'actual emissions' after the change must be assumed to be equal to 'potential to emit'"); Ex. 10, Excerpt of EPA New Source Review Workshop Manual, Prevention of Significant Deterioration and Nonattainment Area Permitting, Draft, October 1990, p. A.41 ("For proposed new or modified units which have not begun normal operations, the potential to emit must be used to determine the increase from the units.").

The federal regulations thus state that for any emissions unit that "has not begun normal operations," the unit's post-change emissions shall equal its "potential to emit." 40 C.F.R. § 52.21(b)(21)(iv) (1997). EPA interprets this provision to apply to all new units and all existing units that will undergo a non-exempt physical or operational change. This interpretation is not clearly erroneous or inconsistent with the regulations. A new unit has not begun operations at all. An existing unit that will undergo a non-exempt physical or operational change (other than a "like kind replacement")[10] has no operating history in its altered state. It is, therefore, reasonable

[10] The Seventh Circuit, in *Wis. Elec. Power Co. v. Reilly*, 893 F.2d 901 (7th Cir. 1990), created an exception to the actual to potential test for "like kind replacements" undertaken at electric

for EPA – and the State of Colorado, with its identical SIP provision – to presume that the new or modified unit will be operated at its maximum capacity. This interpretation is entirely consistent with the federal regulations and the Colorado SIP. Indeed, this interpretation is compelled by their plain language. Accordingly, EPA's interpretation is entitled to deference.

(c) Case Law Supports the Use of an "Actual-to-Potential" Test Under the Circumstances of this Case.

Courts have consistently upheld EPA's use of the "actual-to-potential" test to determine whether a modification will result in a net emissions increase of a pollutant under the PSD regulations. For example, the First Circuit upheld the use of the test to determine PSD applicability for a cement kiln project. *Puerto Rican Cement Co. v. EPA*, 889 F.2d 292 (1st Cir. 1989). The project involved converting an existing kiln (kiln no. 6) from a "wet" to a "dry" process and combining it with another existing kiln (kiln no. 3). At the time of the project, kilns 3 and 6 were operating at 60% of their combined capacity. The converted kiln would have more capacity than the combined capacities of kilns 3 and 6 but a lower emissions rate. Therefore, at any given production level, the converted kiln would emit less pollution than the two existing kilns. If the converted kiln was operated at full capacity, however, it would produce more pollution than the existing kilns produced when operated at 60% of their capacity. In determining the change in emissions attributable to this project:

> EPA calculated the actual historical amount of pollutants that Kilns 3 and 6 emitted in the past (which, under the regulations, equals the average emissions over the past two years), and compared that with the amount of pollutants that the converted kiln would be capable of emitting in the future. Since the Company

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utilities. This exception does not apply here. The modifications at the Lyons Facility were not a "like kind replacement" projects, and the Facility is not an electric utility.

> operated the kilns at only 60 percent of their capacity in 1985-86, the new kiln, though cleaner and more efficient, is obviously capable of emitting significantly more pollutants.

Id. at 296. The company challenged EPA's use of the actual-to-potential test, calling it improper, arbitrary, and inconsistent with the regulations. The First Circuit disagreed. The court found that the actual-to-potential test applied because the modified emissions unit had "not begun normal operation." *Id.* at 297. The court further found that the "language and expressed intent of the regulations both support[ed] EPA's interpretation." *Id.* at 296.

In *United States v. Murphy Oil USA, Inc.*, 143 F. Supp. 2d 1054 (W.D. Wisc. 2001), the court had to determine the proper emissions test to use in connection with a project involving an existing sulfur recovery unit at an oil refinery:

> In 1987 and 1988, defendant made physical changes to the sulfur recovery unit at a cost of more than $250,000 by upgrading the primary burner, replacing the catalyst, installing a new and larger combustion chamber, adding an acid gas knock-out drum and a tail gas analyzer, as well as adding a hydrocarbon coalescer in the upstream amine unit and installing a distributed control system. These changes served to improve the reliability of the unit and to improve its sulfur recovery efficiency.

Id. at 1066. The court explained that the relevant inquiry was whether these changes to the existing sulfur recovery unit "were sufficiently significant to support a finding that normal operations had not begun before the improvements." *Id.* at 1104. The court went on to state that "[i]f normal operations had not begun, an actual-to-potential test applies; otherwise, an actual-to-future-actual prediction is appropriate." *Id.* Applying a literal definition of "normal operations," the court found that the changes to the sulfur recovery unit were "significant enough to make the post-construction unit effectively a new unit that had not begun normal operations at the start of construction." *Id.* at 1105. Of particular importance to the court was the fact that the changes did not simply involve replacing old parts with equivalent new ones. *Id.* The court concluded by noting that courts give substantial deference to an agency's interpretation of its own regulations: "I cannot say that [EPA's] interpretation of the applicable test is unreasonable. I conclude that the "actual-to-potential" test is appropriate to determine whether defendant is subject to [PSD] standards." *Id.*

In *Sierra Club v. Morgan*, No. 07-C-251-S, 2007 WL 3287850 (W.D. Wis. Nov. 7, 2007), the court considered how to assess the emissions impact of modifications to several existing coal-fired boilers used to provide building heat. The court found that four projects (numbered 2 through 5) were non-exempt physical changes and that this fact gave rise to "the presumption that the projects were sufficiently significant to support a finding that normal operations had not begun before the physical changes occurred." *Id.* at *19. The court then analyzed each of the four projects. Other than the first (project 2), which "involved the changing of old tubes with new tubes without any change in the design or functionality," the court found that each project sufficiently changed the units so that it could not be said that they had begun normal operations before the changes. For these projects, the court determined that emissions changes should be measured using the "actual to potential" test. *Id.*

Thus, courts assessing net emissions increases that result from a modification under the PSD regulations have consistently found that "normal operations" had not begun prior to the modifications, except where the change merely reflects the replacement of old parts with equivalent new ones. It is undisputed that the changes at the Lyons Facility went beyond the replacement of old parts. As discussed above, CEMEX has admitted to substantive changes to the Raw Mill, Kiln System, and Finish Mill in the period beginning in 1997. Accordingly, it had not "begun normal operations" prior to the modification, and therefore the "actual-to-potential" emissions comparison should apply for the purpose of determining whether there was a significant net increase in emissions.

5. Element 6: CEMEX Did Not Obtain a PSD Permit for the Modification.

CEMEX does not dispute that it did not obtain a PSD permit prior to the 1997 modification. In response to requests to admit, CEMEX admitted that it did not seek or obtain a construction permit for the combustion chamber change, the use of liquid oxygen, or the construction of an oxygen plant, among other things. *See, e.g*, Ex. 3, Excerpt of Defendant CEMEX, Inc.'s Second Amended Responses to United States' First Set of Requests for Admissions, Response Nos. 40, 103, and 140, pp. 19, 49, and 64.

CONCLUSION

Partial summary judgment is appropriate here because there is no genuine dispute of material fact. For that reason, the United States requests that this Court find: (1) that the Lyons Facility had not begun normal operations prior to modifying the Facility in 1997; and (2) that the appropriate emissions comparison to determine whether there is a net emissions increase under the Colorado SIP is, therefore, the pre-modification actual emissions verses the post-modification potential to emit.

Respectfully Submitted,

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CERTIFICATE OF SERVICE

I hereby certify that on November 14, 2011, the foregoing United States' Motion for Partial Summary Judgment was filed electronically using the Court's ECF system and automatically served on counsel of record, identified below:

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